INTERNATIONAL BACCALAUREATE

Mathematics: analysis and approaches

Math AA

EXERCISES [Math-AA 2.4-2.5]

COMPOSITION – INVERSE FUNCTION

O. Practice questions

1. [Maximum mark: 10] ***[without GDC]***

Let $f(x) = 10 - 2x$ and $g(x) = 5x$. Calculate

(a) $(f \circ g)(x)$ and $(g \circ f)(x)$ [2]

(b) $f^{-1}(x)$ [2]

(c) $g^{-1}(10)$ [2]

(d) $(f^{-1} \circ g)(x)$ and $(g \circ f)^{-1}(x)$ [2]

(e) $(f \circ f)(x)$ and $(g \circ g)(x)$ [2]

2. [Maximum mark: 6] ***[without GDC]***

Complete the following table with the inverse of each function

Original function	Inverse function
$f(x) = x + 5$	$f^{-1}(x) = x - 5$
$f(x) = x - 5$	
$f(x) = x + 100$	
$f(x) = 3x$	
$f(x) = \frac{x}{5}$	
$f(x) = x^3$	
$f(x) = 3x + 100$	

3. [Maximum mark: 4] ***[without GDC]***

Consider the following table of values for f

x	1	2	3	4	5	6
$f(x)$	3	4	5	6	1	2

(a) Write down the values of

(i) $f(1)$ (ii) $f^{-1}(1)$ [2]

(b) Given that $f(x)=2$ write down the value of x. [1]

(c) Given that $f^{-1}(x)=2$ write down the value of x. [1]

4. [Maximum mark: 8] ***[without GDC]***

The tables below show some values of two functions f and g

x	1	2	3	4
$f(x)$	2	-3	-1	3

x	1	2	3	4
$g(x)$	5	3	1	-3

(a) Write down the values of $g(3)$, $f^{-1}(3)$. [2]

(b) Calculate $(f\circ g)(2)$ [2]

(c) Calculate $(g\circ g)(3)$ [2]

(d) Find a solution of the equation $(g\circ f)(x)=3$. [2]

5. [Maximum mark: 5] ***[without GDC]***

Find the inverse of the function $f(x)=\dfrac{x}{x+5}$

6. [Maximum mark: 6] ***[without GDC]***

Let $f(x)=2^x$, and $g(x)=\dfrac{x}{x-2}$, $x\neq 2$. Find

(a) $(g\circ f)(3)$; [3]

(b) $g^{-1}(5)$. [3]

METHOD A [by finding $(g\circ f)(x)$ and $g^{-1}(x)$ first]

METHOD B [directly by definitions: $(g\circ f)(3)=g(f(3))$ and $g^{-1}(5)=x \Leftrightarrow g(x)=5$]

<u>Remark:</u> Confirm the result 6(b) by using **GDC - Graph – [G-Solv] – [X-CAL]**

7. [Maximum mark: 10] ***[without GDC]***

The function f is given by $f(x) = x^2$, for $0 \le x \le 3$.

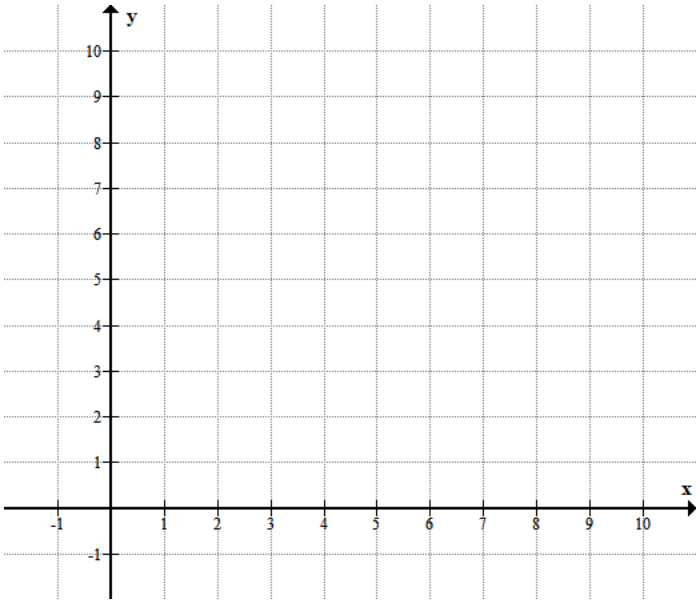
(a) Sketch the graph of f. [2]

(b) State the domain and the range of f. [2]

(c) Find the inverse function f^{-1}. [2]

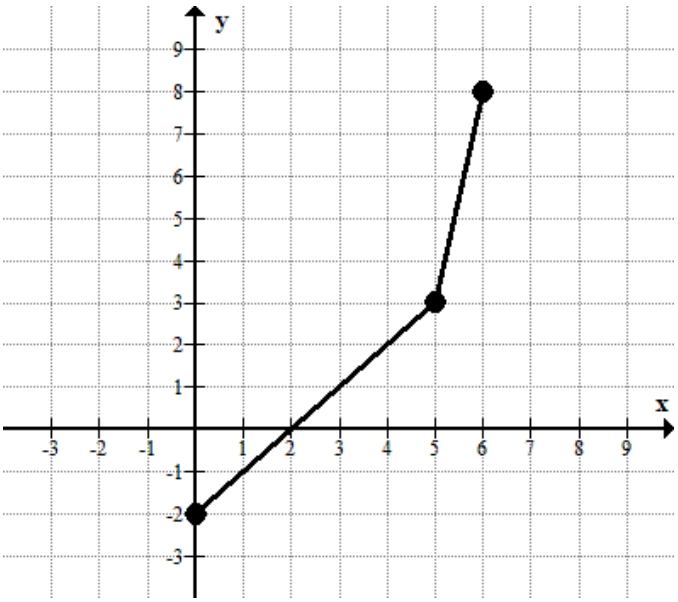
(d) On the same axes with f, sketch the graph of f^{-1} [2]

(e) State the domain and the range of f^{-1}. [2]



8. [Maximum mark: 12] ***[without GDC]***

The graph of the function f is shown below.



(a) Write down the values of

(i) $f(0)$ (i) $f(2)$ (ii) $f(4)$ [3]

(b) Write down the values of

(i) $f^{-1}(3)$ (ii) $f^{-1}(8)$ (iii) $f^{-1}(-1)$ (iv) $f^{-1}(0)$ [4]

(c) Write down the solution of the equation $f(x)=0$ [1]

(d) Write down the solution of the equation $f(x)=2$ [1]

(e) On the diagram above sketch the graph of f^{-1} [3]

9*. [Maximum mark: 8] ***[without GDC]***

Let $f(x)=\dfrac{2x-3}{3x-2}$. Show that

(a) $f^{-1}=f$ (i.e. the function is self-inverse) [4]

(b) $(f\circ f)(x)=x$ [4]

10*. [Maximum mark: 7] ***[without GDC]***

The function f is defined by $f:x\mapsto x^3$.

(a) Find an expression for f^{-1}. [2]

(b) Find an expression for $g(x)$ in terms of x in each of the following cases

(i) $(f\circ g)(x)=x+1$;

(ii) $(g\circ f)(x)=x+1$. [5]

11.** [Maximum mark: 6] ***[without GDC]***

(a) Given that $f\circ g=h$, express f in terms of g and h. [2]

(b) Given that $f\circ g=h$, express g in terms of f and h. [2]

(c) Given that $f\circ g\circ h=k$, express g in terms of f, h and k. [2]

A. Exam style questions (SHORT)

12. [Maximum mark: 6] ***[without GDC]***

Let $g(x)=3x-2$, $h(x)=\dfrac{5x}{x-4}$, $x\neq 4$.

(a) Find an expression for $(h\circ g)(x)$. Simplify your answer. [3]

(b) Solve the equation $(h\circ g)(x)=0$. [3]

13. [Maximum mark: 6] ***[without GDC]***

The functions f and g are defined by $f:\mapsto 3x,\ g:x\mapsto x+2$.

(a) Find an expression for $(f\circ g)(x)$. [2]

(b) Show that $f^{-1}(18)+g^{-1}(18)=22$. [4]

14. [Maximum mark: 6] ***[without GDC]***

Let $f(x) = 2x + 1$ and $g(x) = 3x^2 - 4$. Find

(a) $f^{-1}(x)$; [2]

(b) $(g \circ f)(-2)$; [2]

(c) $(f \circ g)(x)$. [2]

15. [Maximum mark: 4] ***[without GDC]***

The function f is defined by $f: x \mapsto \sqrt{3 - 2x}$, $x \le \frac{3}{2}$

Evaluate $f^{-1}(5)$.

16. [Maximum mark: 6] ***[without GDC]***

Let $f(x) = \sqrt{x + 4}$ $x \ge -4$ and $g(x) = x^2$, $x \in \mathbb{R}$.

(a) Find $(g \circ f)(3)$ [2]

(b) Find $f^{-1}(x)$ [3]

(c) Write down the domain of f^{-1}. [1]

17. [Maximum mark: 5] ***[without GDC]***

Let $f(x) = x^3 - 4$ and $g(x) = 2x$.

(a) Find $(g \circ f)(-2)$. [2]

(b) Find $f^{-1}(x)$ [3]

18. [Maximum mark: 6] ***[without GDC]***

Consider the functions $f(x) = 2x$ and $g(x) = \frac{1}{x - 3}$, $x \ne 3$.

(a) Calculate $(f \circ g)(4)$. [2]

(b) Find $g^{-1}(x)$ [3]

(c) Write down the domain of g^{-1}. [1]

19. [Maximum mark: 4] ***[without GDC]***

Two functions f, g are defined as follows:

$$f: x \mapsto 3x + 5 \qquad\qquad g: x \mapsto 2(1 - x)$$

Find

(a) $f^{-1}(2)$; [2]

(b) $(g \circ f)(-4)$. [2]

20. [Maximum mark: 6] ***[without GDC]***

Two functions f and g are defined as follows:

$$f(x) = \cos x, \qquad g(x) = 2x + 1$$

(a) Find $(g \circ f)(x)$. [2]

(b) Find $(f \circ g)(x)$. [2]

(c) Find $(g^{-1} \circ g)(x)$ and $(g \circ g^{-1})(x)$. [2]

21. [Maximum mark: 6] ***[without GDC]***

Consider the functions $f : x \mapsto 4(x-1)$ and $g : x \mapsto \dfrac{6-x}{2}$.

(a) Find g^{-1}. [2]

(b) Solve the equation $(f \circ g^{-1})(x) = 4$. [4]

22. [Maximum mark: 5] ***[without GDC]***

Let $f(x) = x^2$ and $g(x) = 2x - 3$.

(a) Find $g^{-1}(x)$. [2]

(b) Find $(f \circ g)(4)$. [3]

23. [Maximum mark: 5] ***[without GDC]***

Let $f(x) = 7 - 2x$ and $g(x) = x + 3$.

(a) Find $(g \circ f)(x)$. [2]

(b) Write down $g^{-1}(x)$. [1]

(c) Find $(f \circ g^{-1})(5)$. [2]

24. [Maximum mark: 5] ***[without GDC]***

Let $f(x) = 3x$, $g(x) = 2x - 5$ and $h(x) = (f \circ g)(x)$.

(a) Find $h(x)$. [2]

(b) Find $h^{-1}(x)$. [3]

25. [Maximum mark: 6] ***[with GDC]***

Let $f(x) = \sqrt{x}$, and $g(x) = 2^x$.

(a) Find $(f^{-1} \circ g)(x)$ [4]

(b) Solve the equation $(f^{-1} \circ g)(x) = 16$ [2]

26. [Maximum mark: 6] ***[without GDC]***

Consider the functions f and g where $f(x) = 3x - 5$ and $g(x) = x - 2$.

Solve the equation $(f^{-1} \circ g)(x) = (g^{-1} \circ f)(x)$.

27. [Maximum mark: 6] ***[without GDC]***

Let $f(x)=\dfrac{4}{x+2}$, $x\neq -2$ and $g(x)=x-1$. If $h=g\circ f$, find

(a) $h(x)$ [2]

(b) $h^{-1}(x)$ [4]

28. [Maximum mark: 4] ***[without GDC]***

Given functions $f:x\mapsto x+1$ and $g:x\mapsto x^3$, find the function $(f\circ g)^{-1}$

29. [Maximum mark: 5] ***[without GDC]***

If $f(x)=\dfrac{x}{x+1}$, for $x\neq -1$ and $g(x)=(f\circ f)(x)$, find

(a) $g(x)$. [3]

(b) $(g\circ g)(2)$. [2]

30. [Maximum mark: 5] ***[without GDC]***

The function f is defined as $f(x)=\dfrac{3x-4}{x+2}$, $x\neq -2$.

(a) Find an expression for $f^{-1}(x)$. [4]

(b) Write down the domain of f^{-1}. [1]

31*. [Maximum mark: 9] ***[with / without GDC]***

The function f is given by $f(x)=x^2-6x+13$, for $x\geq 3$.

(a) Show that f may also be written in the form $f(x)=(x-3)^2+4$. [2]

(b) **Hence** find the inverse function f^{-1}. [3]

(c) State the domain and the range of f. [2]

(d) State the domain and the range of f^{-1}. [2]

32*. [Maximum mark: 7] ***[without GDC]***

The functions f and g are both defined in the interval [-4,4] and g is invertible.

Some values of the functions are given below.

x	1	3	4
$f(x)$	3	2	1

x	1	3	4
$g(x)$	4	1	-3

(a) Calculate $(f\circ f)(1)$. [2]

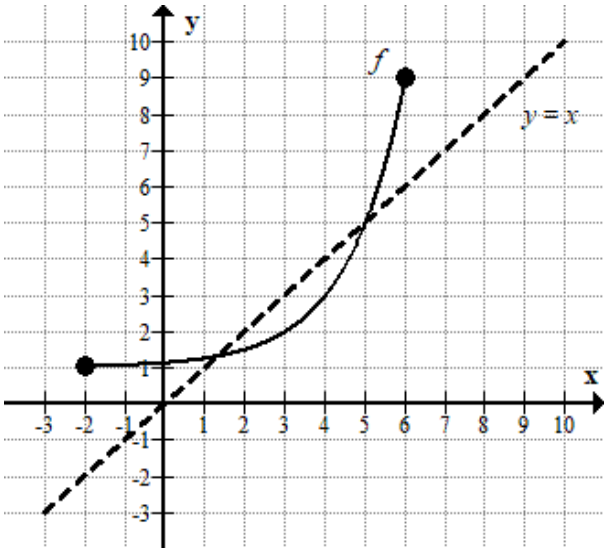
(b) Calculate $(g^{-1}\circ f)(4)$. [2]

(c) Find a solution of the equation $(f\circ g)(x)=1$. [2]

(d) Calculate $(g^{-1}\circ g)(2)$. [1]

33. [Maximum mark: 4] ***[without GDC]***

The graph of the function f is shown below. On the same diagram sketch the graph of the function f^{-1}.



34*. [Maximum mark: 6] ***[with / without GDC]***

The functions $f(x)$ and $g(x)$ are given by $f(x)=\sqrt{x-2}$ and $g(x)=x^2+x$.

The function $(f\circ g)(x)$ is defined for $x\in\mathbb{R}$, **except** for the interval $]a,b[$

(a) Calculate the value of a and of b. [5]

(b) Find the range of $f\circ g$. [1]

35.** [Maximum mark: 5] ***[with GDC]***

Consider the functions

$$f(x)=3x^2-12x+7,\ x\geq a \quad \text{and} \quad g(x)=3x^2-12x+7,\ x\leq b.$$

(a) Given that f^{-1}, g^{-1} exist, find

(i) the smallest possible value of a. (ii) the largest possible value of b. [3]

(b) Within the domains determined in (a), find

(i) $f^{-1}(7)$. (ii) $g^{-1}(7)$ [2]

36*. [Maximum mark: 6] ***[without GDC]***

The function f is defined by $f:x\mapsto x^3-1$

(a) Find $g(x)$ given that $(f\circ g)(x)=2x+1$; [3]

(b) Find $g(x)$ given that $(g\circ f)(x)=2x+1$. [3]

37.** [Maximum mark: 6] ***[without GDC]***

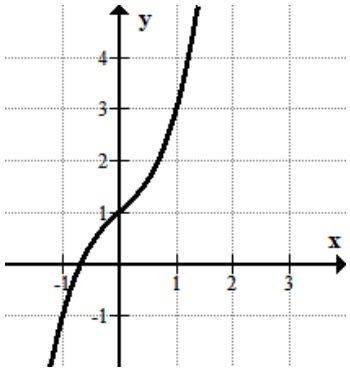
Let f and g be two functions. Given that $(f \circ g)(x) = \frac{x+1}{2}$ and $g(x) = 2x - 1$, find $f(x-3)$.

38*. [Maximum mark: 6] ***[without GDC]***

The function f is defined for $x \le 0$ by $f(x) = \frac{x^2 - 1}{x^2 + 1}$. Find an expression for $f^{-1}(x)$.

39.** [Maximum mark: 6] ***[with GDC]***

The following diagram shows part of the graph of $f(x) = x^3 + x + 1$



(a) Find (i) $f^{-1}(1)$ (ii) $f^{-1}(0)$ (iii) $f^{-1}(2)$. [3]

(b) Solve $f(x) = f^{-1}(x)$. [3]

B. Exam style questions (LONG)

40. [Maximum mark: 12] ***[with / without GDC]***

The function f is given by $f(x) = 2x^2 - 8x + 1$ for $x \le 2$.

(a) Find $(f \circ f)(0)$. [2]

(b) Express the function in the form $f(x) = a(x-h)^2 + k$. [3]

(c) State the range of f. [1]

(d) Find the inverse function f^{-1}. [4]

(e) State the domain and the range of f^{-1}. [2]

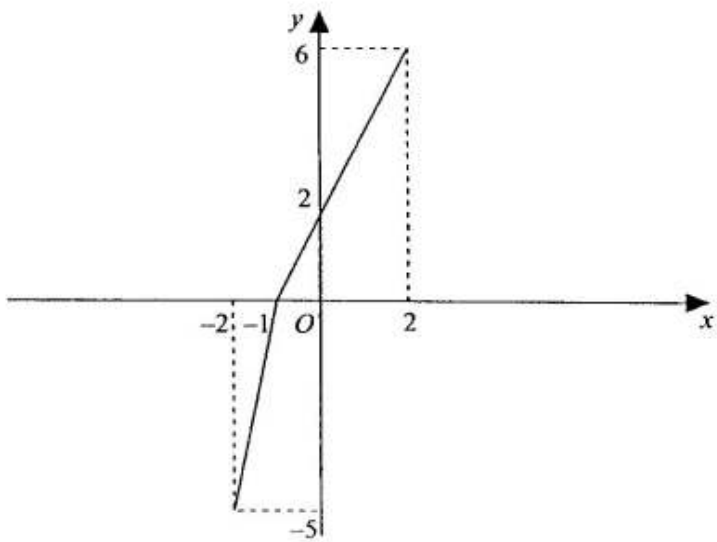
41. [Maximum mark: 10] ***[without GDC]***

The function $f(x)$ is defined by $f(x)=\dfrac{3x-1}{x-3}$, $x \neq 3$.

(a) Show that f is a self-inverse function, that is $f^{-1}=f$ [4]

(b) **Hence** find, in terms of *k*, the value of $(f \circ f)(k)$, where $k \neq 3$. [1]

The figure below shows a sketch of a one-to-one function $g(x)$ defined over the domain $-2 \leq x \leq 2$. The graph of $y=g(x)$ consists of two straight line segments and the range of $g(x)$ is $-5 \leq g(x) \leq 6$.



(c) Find the value of $(f \circ g)(-2)$. [2]

(d) On the same diagram above, sketch the graph of the inverse function $y=g^{-1}(x)$ and state its domain. [3]

42.** [Maximum mark: 13] ***[without GDC]***

(a) The function f is defined by $f(x)=(x+2)^2-3$.

The function g is defined by $g(x)=ax+b$, where a and b are constants.

Find the value of a, $a>0$ and the corresponding value of b, such that

$$f\left(g(x)\right)=4x^2+6x-\frac{3}{4}.$$ [8]

(b) The functions h and k are defined by $h(x)=5x+2$ and $k(x)=cx^2-x+2$ respectively. Find the value of c are such that $h\left(k(x)\right)$ has equal roots. [5]

43.** [Maximum mark: 10] ***[without GDC]***

Let $f(x)=\sqrt{x+1}+1$ and $g(x)=x^2$.

(a) Solve the equation $(g\circ f)(x)=1$. [3]

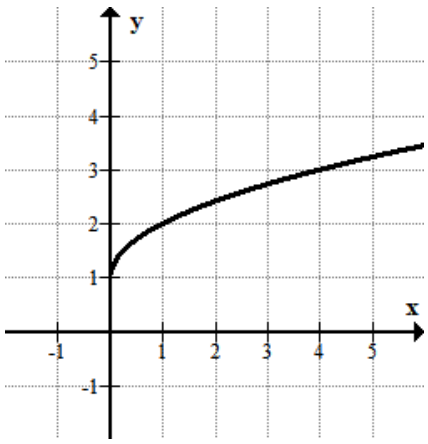
(b) Find f^{-1} in a simplified form. [3]

(c) Find the function $h(x)$ given that $h\circ f=g$. [2]

(d) Find the function $k(x)$ given that $f\circ k=g$. [2]

44.** [Maximum mark: 11] ***[without GDC]***

The following diagram shows part of the graph of the function f is defined by $f(x)=1+\sqrt{x}$, for $x\geq 0$



(a) Find an expression for $f^{-1}(x)$. [3]

(b) State the domain and the range of $f^{-1}(x)$. [2]

(c) On the same diagram above sketch the graph of $f^{-1}(x)$. [2]

(d) Find the **exact** value of the solution of the equation $f(x)=f^{-1}(x)$. [4]